Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The point of intersection of the normals to the parabola $y^2 = 4x$ at the ends of its latus rectum is

(A) $(0, 2)$

(B) $(3, 0)$

(C) $(0, 3)$

(D) $(2, 0)$

Q.2

Equation of a normal to the parabola $y^2 = 32x$ passing through its focus is

(A) $x = 0$

(B) $y = 0$

(C) $x + y - 8 = 0$

(D) $x - y - 8 = 0$

Q.3

If the common tangent in the first quadrant of the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$ meets the coordinate axes at A and B, the length of AB is

(A) $7/\sqrt{3}$ units

(B) $14/\sqrt{3}$ units

(C) $4\sqrt{7}$ units

(D) 14 units

Q.4

Equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is

(A) $\sqrt{3}y = 3x + 1$

(B) $\sqrt{3} \quad y = -(x + 3)$

(C) $\sqrt{3}y = x + 3$

@IITJEE_Advanced

(D) $\sqrt{3}y = -(3x+1)$

Q.5

An equation of a common tangent to the parabola $y^2 = 4x$ and $x^2 = 4y$ is

(A) $x - y + 1 = 0$

(B) $x + y - 1 = 0$

(C) $x + y + 1 = 0$

(D) $y = 0$

Q.6

The normal at the point $\left(bt_1^2, 2bt_1\right)$ on a parabola meets the parabola again in the point $\left(bt_2^2, 2bt_2\right)$, then

(A) $t_2 = t_1 + \frac{2}{t_1}$

(B) $t_2 = -t_1 - \frac{2}{t_1}$

(C) $t_2 = -t_1 + \frac{2}{t_1}$

(D) $t_2 = t_1 - \frac{2}{t_1}$

Q.7

Consider the normals drawn at three different points on the parabola $y^2 = 4x$ passing through the point h, k, then

(A) $h < 2$

(B) $h > 2$

(C) $h < 3$

(D) $h > 3$

Q.8

Let P be any point on the parabola $y^2 = 4ax$ whose focus is S. If normal at P meet x -axis at Q then ΔPSQ is always

(A) isosceles

(B) equilateral

(C) right angled

(D) None of these

Q.9

@IITJEE_Advanced

The equation of one of the common tangents to the parabola $y^2 = 8x$ and $x^2 + y^2 - 12x + 4 = 0$ is

(A) $y = -x + 2$

(B) $\mathrm{y} = \mathrm{x} - 2$

(C) $y = x + 2$

(D) None of these

Q.10

The equation of a tangent to the parabola $y^2 = 5x$ is $x + y = 10$. If from the point (m,n) on this tangent the other tangent drawn to the parabola is perpendicular to the given tangent, then the value of n - 5 m is

(A) 17.5

(B) 18.5

(C) 19.5

(D) 20.5

Q.11

If tangents are drawn from any point on the line $x + 4a = 0$ to the parabola $y^2 = 4ax$, then their chord of contact subtends angle at the vertex equal to

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{6}$

Q.12

The locus of the point of intersection of two tangents to the parabola $y^2 = 4ax$, which are at right angle to one another is

(A) $x^2 + y^2 = a^2$

(B) $ay^2 = x$

(C) $x + a = 0$

(D) $x + y \pm a = 0$

Q.13

If tangents are drawn to the parabola $y^2 = 4ax$ at points whose abscissae are in the ratio $\mathrm{m}^2 : 1$, then the locus of their point of intersection is the curve $(\mathrm{m} > 0)$.

(A) $y^2 = \left(m^{1/2} - m^{-1/2}\right)^2 ax$

@IITJEE_Advanced

(B) $y^2 = \left(m^{1/2} + m^{-1/2}\right)^2 ax$

(C) $y^2 = \left(m^{1/2} + m^{-1/2}\right)^2 x$

(D) None of these

Q.14

Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points $(2, 0)$ and $(3, 0)$ is

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{6}$

Q.15

The locus of the point of intersection of the perpendicular tangents to the parabola $x^2 = 4xy$, is

(A) $y = a$

(B) $y = -a$

(C) $x = a$

(D) $x = -a$

Answers & Solutions

Q.1 Answer:

$(3, 0)$

Solution:

Any point on the parabola is $\left(at^2, 2at\right)$ **where** $a = 1$**. For the latus rectum** $x = a \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$**. So ends of latus rectum are** $(1, \pm 2)$**. Equations of the normals at these points is given by**

$y = -tx + 2at + at^3$

$\Rightarrow \quad x + y - 3 = 0$ **and** $x - y - 3 = 0$

which intersect at $(3, 0)$**.**

Q.2 Answer:

$y = 0$

Solution:

Equation of a normal to the parabola is $y = mx - 2am - am^3, 4a = 32 \Rightarrow a = 8$

which passes through the focus $(8, 0)$ **if** $0 = 8m - 16m - 8m^3 = 0$

$\Rightarrow m^3 + m = 0 \Rightarrow m\left(m^2 + 1\right) = 0 \Rightarrow m = 0$

and the required equation of the normal is $y = 0$**.**

@IITJEE_Advanced

Q.3 Answer:

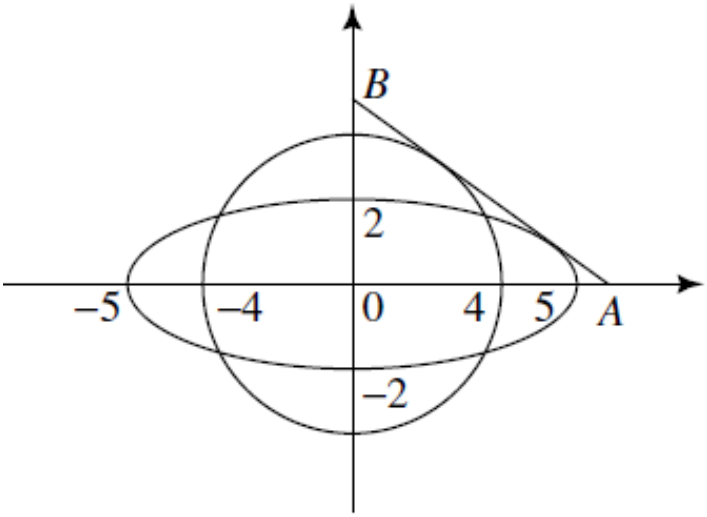
$14/\sqrt{3}$ units

Solution:

An equation of the tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{4}=1$ **is**

$y=mn+\sqrt{25m^2+4}$ **(1)**

As the tangent is in the first quadrant $m<0$.



The line will touch the circle $x^2+y^2=16$ **if** $\frac{m(0)-0+\sqrt{25m^2+4}}{\sqrt{m^2+1}}=4$

$\Rightarrow \quad 25m^2+4=16\left(m^2+1\right)$

$\Rightarrow \quad m^2=\frac{4}{3}\Rightarrow m=\frac{-2}{\sqrt{3}}$

So equation of the tangent is

$y=\frac{-2}{\sqrt{3}}x+\sqrt{25\left(\frac{4}{3}\right)+4}$

$\Rightarrow \quad 2x+\sqrt{3}y=4\sqrt{7}$

which meets the coordinate axes at $A\left(\frac{4\sqrt{7}}{2},0\right)$ **and** $B\left(0,\frac{4\sqrt{7}}{\sqrt{3}}\right)$

Q.4 Answer:

$\sqrt{3}y=x+3$

Solution:

Equation of a tangent to the parabola $y^2=4x$ **is** $y=mx+1/m$**. It will touch the circle** $(x-3)^2+y^2=9$ **whose centre is (3, 0) and radius is 3 if**

$\left|\frac{0+m(3)+(1/m)}{\sqrt{1+m^2}}\right|=3$

or if $(3m+1/m)^2=9\left(1+m^2\right)$

or if $9m^2+6+1/m^2=9+9m^2$

or if $m^2=1/3$ **i.e.** $m=\pm 1/\sqrt{3}$

As the tangent is above the x**-axis, we take** $m=1/\sqrt{3}$ **and thus the required equation is** $\sqrt{3}y=x+3$.

@IITJEE_Advanced

Q.5 Answer:

$x + y + 1 = 0$

Solution:

Equation of a tangent to $y^2 = 4x$ is $y = mx + 1/m$. If it touches the parabola $x^2 = 4y$ then the roots of the equation $x^2 = 4(mx + 1/m)$ are coincident. So $(4m)^2 + \frac{16}{m} = 0 \Rightarrow m = -1$ and the equation of the tangent is $x + y + 1 = 0$

Q.6 Answer:

$t_2 = -t_1 - \frac{2}{t_1}$

Solution:

Equation of the normal to a parabola $y^2 = 4bx$ at point $\left(bt_1^2, 2bt_1\right)$ is $y = -t_1x + 2bt_1 + bt_1^3$
As given, it also passes through $\left(bt_2^2, 2bt_2\right)$ then $2bt_2 = -t_1bt_2^2 + 2bt_1 + bt_1^3$
$\Rightarrow 2 = -t_1\left(t_2 + t_1\right) \Rightarrow t_2 + t_1 = -\frac{2}{t_1}$
$\Rightarrow t_2 = -t_1 - \frac{2}{t_1}$

Q.7 Answer:

$h > 2$

Solution:

Let the normals be
$= mx - 2m - m^3$
These pass through h, k
$k = mh - 2am - am^3$
The three roots m_1, m_2 **and** m_3 **of this cubic on the slope of the three normals. Taking** $a = 1,$ **we get**
$m^3 + 2 - hm + k = 0$
Sum of roots, $m_1 + m_2 + m_3 = 0 \ldots .$ i
$m_1m_2 + m_2m_3 + m_3m_1 = 2 - h \ldots .$ ii
$m_1m_2m_3 = -k \ldots .$ iii **[As** m_1, m_2, m_3 **are real** $m_1^2 + m_2^2 + m_3^2 > 0$ **and not all zero]**
$\Rightarrow m_1 + m_2 + m_3{}^2 - 2$
$m_1m_2 + m_2m_3 + m_3m_1 > 0$
$0 - 22 - h > 0 \Rightarrow h > 2$

Q.8 Answer:

isosceles

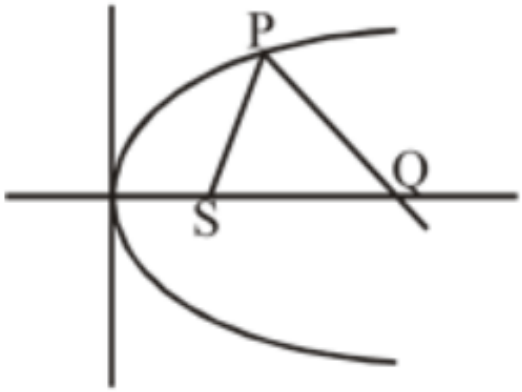
Solution:

Let coondinates of P be $\left(at^2, 2at\right)$, then equation of normal is
$at^3 + (2a - x)t - y = 0 \ldots .(1)$
$\therefore$ PQ **meet** x **-axis at** Q
$\therefore$ **Coordinates of the** Q **is** $\left(at^2 + 2a, 0\right)$**. Cocrdinates of** S **is** $(a, 0)$

@IITJEE_Advanced

$\therefore \mathrm{SQ} = at^2 + a = c\left(1 + t^2\right)$ **and** $\mathrm{PS} = a\sqrt{\left(t^2 - 1\right)^2 + 4t^2} = a\left(1 + t^2\right) = QS$

$\therefore$ PSQ is isosceles.



Q.9 Answer:

$y = x + 2$

Solution:

Any tangent to parabola $y^2 = 8x$ is $y = mx + \frac{2}{m}$
It touches the circle $x^2 + y^2 - 12x + 4 = 0$, if the length of perpendicular from the centre $(6, 0)$ is equal to radius $\sqrt{32}$.

$\therefore \frac{6m + \frac{2}{m}}{\sqrt{m^2 + 1}} = \pm\sqrt{32} \Rightarrow \left(3m + \frac{1}{m}\right)^2 = 8\left(m^2 + 1\right)$

$\Rightarrow \left(3\,\mathrm{m}^2 + 1\right)^2 = 8\left(\,\mathrm{m}^4 + \mathrm{m}^2\right)$

$\Rightarrow m^4 - 2m^2 + 1 = 0 \Rightarrow m = \pm 1$

Hence, the required tangents are $\mathrm{y} = \mathrm{x} + 2$ and $y = -x - 2$

Q.10 Answer:

17.5

Solution:

The locus of the point of intersection of perpendicular tangents to a parabola is its directrix.
(Property of parabola)
The directrix of the parabola
$y^2 = 5x$ is $x = -\frac{5}{4}$
Solving $x = -\frac{5}{4}$ and the given equation of tangent we get the point of intersecion as
$-\frac{5}{4}, \frac{45}{4}$
$\Rightarrow m = -\frac{5}{4}, n = \frac{45}{4}$
$\Rightarrow n - 5m = \frac{45 + 25}{4} = \frac{70}{4} = 17.5$

Q.11 Answer:

$\frac{\pi}{2}$

Solution:

@IITJEE_Advanced

Let $\mathrm{R}(-4\mathrm{a}, \mathrm{k})$ be any point on the line $x = -4a$. The equation of chord of contact PQ w.r.t. $\mathrm{P}(-4\mathrm{a}, \mathrm{k})$ is

$\mathrm{y} \cdot \mathrm{k} = 2\mathrm{a}(\mathrm{x} - 4\mathrm{a})$...(1)

Making equation of parabola $\mathrm{y}^2 = 4\mathrm{ax}$ homogeneous using (1), we get $\mathrm{y}^2 = 4\mathrm{ax}\left(\frac{2\mathrm{ax}-\mathrm{yk}}{8\mathrm{a}^2}\right)$

$\Rightarrow 8\mathrm{a}^2\mathrm{x}^2 - 8\mathrm{a}^2\mathrm{y}^2 - 4\mathrm{akxy} = 0$

This represents the pair of straight lines AP and AQ. Since coefficient of x^2+ coefficient of $\mathrm{y}^2 = 0 \therefore \angle \mathrm{PAQ} = 90^\circ$ i.e. chord of contact PQ subtends a right angle at the vertex.

Q.12 Answer:

$x + a = 0$

Solution:

Let the two tangents to the parabola $y^2 = 4$ ax be PT and QT which are at right angle to one another at $\mathrm{T}(\mathrm{h}, \mathrm{k})$. Then we have to find the locus of $\mathrm{T}(\mathrm{h}, \mathrm{k})$

We know that $\mathrm{y} = \mathrm{mx} + \frac{\mathrm{a}}{\mathrm{m}}$, where m is the slope is the equation of tangent to the parabola $y^2 = 4ax$ for all m

Since this tangent to the parabola will pass through $\mathrm{T}(\mathrm{h}, \mathrm{k})$ so

$\mathrm{k} = \mathrm{mh} + \frac{\mathrm{a}}{\mathrm{m}}$; or $\mathrm{m}^2\ \mathrm{h} - \mathrm{mk} + \mathrm{a} = 0$

This is a quadratic equation in m so will have two roots, say m_1 and m_2, then $\mathrm{m}_1 + \mathrm{m}_2 = \frac{\mathrm{k}}{\mathrm{h}}$, and $\mathrm{m}_1 : \mathrm{m}_2 = \frac{\mathrm{a}}{\mathrm{h}}$

Given that the two tangents intersect at right angle so $\mathrm{m}_1 \cdot \mathrm{m}_2 = -1$ or $\frac{\mathrm{a}}{\mathrm{h}} = -1$ or $\mathrm{h} + \mathrm{a} = 0$

@IITJEE_Advanced

The locus of $\mathrm{T(h,k)}$ is $\mathrm{x+a=0}$, which is the equation of directrix.

Q.13 Answer:

$y^2 = \left(m^{1/2} + m^{-1/2}\right)^2 ax$

Solution:

Consider two points $\mathrm{P}\left(\mathrm{at_1^2}, \mathrm{2at_1}\right)$ and $\mathrm{Q}\left(\mathrm{at_2^2}, \mathrm{2at_2}\right)$ on the parabola $\mathrm{y^2 = 4ax}$

Given : $\frac{at_1^2}{at_2^2} = \frac{m^2}{1}$ **or** $t_1 = mt_2$

Let $\mathrm{R(h,k)}$ be the point of intersection of tangents at P and Q

Then, $\mathrm{h = at_1t_2}$ **and** $\mathrm{k = a\left(t_1 + t_2\right)}$**...(1)**

$\Rightarrow \quad \mathrm{h = amt_2^2}$ **and** $\mathrm{k = a\left(mt_2 + t_2\right)}$

[Using (1)] $\Rightarrow \quad t_2^2 = \frac{h}{am}$ **and** $t_2 = \frac{k}{a(m+1)}$

Equating the two values of $\mathrm{t_2}$, we $\frac{\mathrm{k^2}}{\mathrm{a^2(\,m+1)^2}} = \frac{\mathrm{h}}{\mathrm{am}}$

$\Rightarrow \quad \mathrm{k^2 = ah\frac{(m+1)^2}{m} \Rightarrow k^2 = ah\left(\sqrt{m} + \frac{1}{\sqrt{m}}\right)^2}$

$\therefore$ **Required locus is** $y^2 = \mathrm{ax}\left(m^{\frac{1}{2}} + m^{-\frac{1}{2}}\right)^2$

Q.14 Answer:

$\frac{\pi}{2}$

Solution:

$\frac{dy}{dx} = 2x - 5$

$\left(\frac{dy}{dx}\right)_{(2,0)} = -1 \quad \left(\frac{dy}{dx}\right)_{(3,0)} = 1$

$\therefore m_1, m_2 = -1 \Rightarrow$ **angle is** $\pi/2$

Q.15 Answer:

$y = -a$

Solution:

Since, the locus of the point of intersection of the $\perp$ tangents to a parabola is its directrix. Therefore, the required locus is

$y = -a.$

@IITJEE_Advanced